Monthly Intelligence Report

Edition 002 — Schleswig-Holstein Wind Corridor

Where the energy transition is outrunning the grid. edition · May 2026 · SSI v4.0.2

CONTENTS

SECTION A

Executive Summary

SUBSTATIONS SCORED	GRID LINES MAPPED	MC SIMULATIONS
13,251 5,973 HV · 7,278 MV · 401 Kreise	**61,625** ~92,000 km coverage	**132.5 M** 10,000 per substation

PUBLIC DATA SOURCES
35 95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 346,000+ source data points per run, executes 132.5 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 2.65 billion arithmetic operations — producing 649,299 output data points with full uncertainty quantification across 13,251 substations spanning 401 Kreise and 61,625 grid lines covering ~92,000 km.

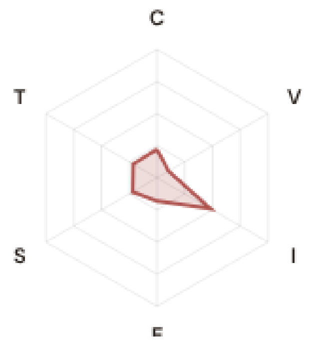
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 35 verified public sources, making the methodology fully auditable and reproducible. Initially covering Italy's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Altriper Straße 25

Heilbronn, Stadt, Baden-Württemberg · 20 kV

0.345	Medium	0.123	4th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

- C Continuity: 0.223 / 0.30 (74%)
- I Infrastructure: 0.481 / 0.25 (192%)
- S Saturation: 0.219 / 0.20 (110%)
- V Voltage: 0.105 / 0.10 (105%)
- E Economic: 0.182 / 0.10 (182%)
- T Transition: 0.214 / 0.05 (428%)

MODIFIERS

- **R3 Consequence:** 1.131 ↑ amplifies
- **R4 Graph Criticality:** 1.125 ↑ amplifies
- **R6a Restoration:** 1.004 → neutral
- **R6b Network Topology:** 1.000 → neutral
- **R7 Digital Readiness:** 1.004 → neutral

This month's spotlight — automatically selected for May 2026 — is **Altriper Straße 25** in Heilbronn, Stadt, Baden-Württemberg. With an R_final of 0.345 (Medium band, 4th fleet percentile), its risk profile is dominated by the T (Transition) component at 428% of its weight ceiling, followed by I (Infrastructure) at 192%. Modifiers collectively shift R_base by 27.5%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

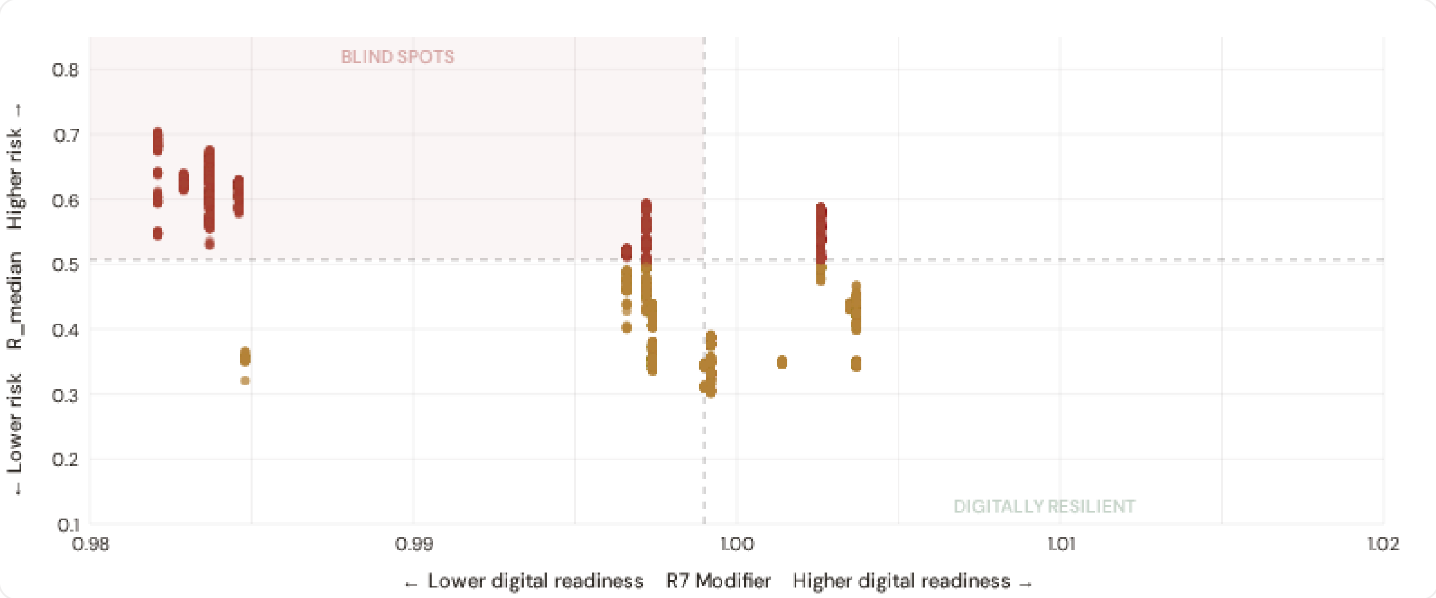
Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Germany's Energiewende transition planning requires.

CYBER HIGH EXPOSURE	BLIND SPOTS	AVG R7 MODIFIER
2372	3559	0.9975
17.9% of fleet	High/Critical + R7 < median	Range [0.99, 1.05]

HIGH-CONSEQUENCE SUBS
6,012
45.4% · R3 ≥ 1.12

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical — Dashed lines = fleet medians

The cyber-physical matrix identifies **3559 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 0.9990). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Brandenburg (671), Rheinland-Pfalz (574), Sachsen (545)**. Across the full fleet, 2372 substations (17.9%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier — indicating strong digital infrastructure coverage but concentrated in the northern urban grid.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 Capital-Intensive / High-Consequence

Est. VoLL: €15–30/kWh

Heavy industry, petrochemical, large manufacturing — highest economic loss per interruption

2,504 substations (18.9%) High/Critical: **1009** (40.3%) Avg R: **0.488** Avg E: **0.210** / 0.10

Low 0 Med 1495 High 1009 Crit 0

🔧 Industrial / Medium-Large Enterprise

Est. VoLL: €8–15/kWh

Diversified industry, logistics, food processing — significant continuity requirements

9,308 substations (70.2%) High/Critical: **5378** (57.8%) Avg R: **0.507** Avg E: **0.221** / 0.10

Low 0 Med 3930 High 5378 Crit 0

🏪 Commercial / SME-Dense

Est. VoLL: €3–8/kWh

Service economy, retail, tourism — moderate individual impact but high aggregate exposure

1,338 substations (10.1%) High/Critical: **581** (43.4%) Avg R: **0.464** Avg E: **0.267** / 0.10

Low 0 Med 757 High 581 Crit 0

🌾 Light / Agricultural / Rural

Est. VoLL: €1–3/kWh

Sparse economic activity, agricultural land — lower VoLL but higher social vulnerability

101 substations (0.8%) High/Critical: **72** (71.3%) Avg R: **0.528** Avg E: **0.235** / 0.10

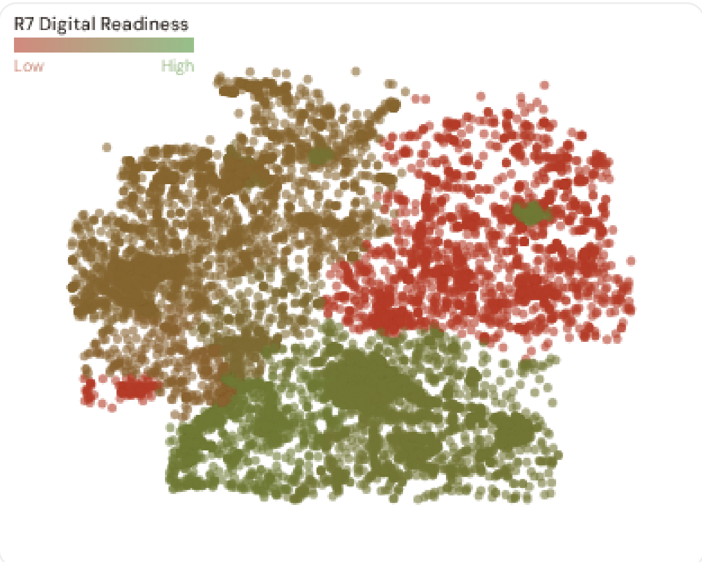
Low 0 Med 29 High 72 Crit 0

Value of Lost Load (VoLL) context: CEER estimates for Germany place average VoLL at €16–22/kWh for industrial customers and €8.5/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **1009 substations** combine capital-intensive economic fabric (R3 ≥ 1.14) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Niedersachsen (481), Nordrhein-Westfalen (457), Sachsen (455)** — regions where industrial corridors depend on uninterrupted power for continuous processes (automotive, chemicals, steel). A single hour of unplanned outage at these substations carries an estimated VoLL of €16–22/kWh for industry, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average unemployment rate is 4.5%, but this masks sharp regional variation: Eastern Bundesländer combine higher unemployment with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Kreis Digital Readiness Ranking

Kreise ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_median to identify Kreise combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 KREISE

Longer bar = higher R7 = better digital infrastructure.

1	Vorpommern-Rügen ⚠ high R	0.9821
2	Landkreis Rostock ⚠ high R	0.9821
3	Rostock, Stadt ⚠ high R	0.9821
4	Ludwigslust-Parchim ⚠ high R	0.9821
5	Schwerin ⚠ high R	0.9821
6	Nordwestmecklenburg ⚠ high R	0.9821
7	Mecklenburgische Seenplatte ⚠ high R	0.9821
8	Vorpommern-Greifswald ⚠ high R	0.9821
9	Magdeburg ⚠ high R	0.9829
10	Halle (Saale) ⚠ high R	0.9829
11	Dessau-Roßlau ⚠ high R	0.9829
12	Ilm-Kreis ⚠ high R	0.9837
13	Barnim ⚠ high R	0.9837
14	Saalfeld-Rudolstadt ⚠ high R	0.9837

15	Potsdam ⚠ high R	0.9837

Kreis-level analysis reveals a clear East-West digital divide mirroring the broader DESI pattern. **Vorpommern-Rügen** has the lowest average R7 (0.9821), while **Esslingen** leads at 1.0037 — a spread of 0.0216 across the R7 range. **58 Kreise** combine below-median digital readiness with above-median grid risk, making them priority targets for Energiewende digitalisation investment. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

> **Inaugural edition — Baseline Snapshot (April 2026).** This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 0.9975, median = 0.9990; 2372 substations classified HIGH cyber-exposure; 3559 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging Kreise where DESI improvements (broadband rollout, smart meter deployment, OT upgrades) translate into measurable R7 shifts. The next material R7 update is expected when Eurostat publishes the 2025 DESI sub-indices (anticipated Q3 2026).

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **35 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

35

95 variables

WEEKLY / MONTHLY

5

High-frequency feeds

QUARTERLY / ANNUAL

22

Regular refresh cycle

STATIC / DERIVED

8

Standards + scenarios

Data Source Registry — May 2026

OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
MASTR	MaStR Bulk Registry (SQLite)	MONTHLY	Kreis	5 vars
TNET	TransnetBW / Amprion / 50Hertz / TenneT	MONTHLY	TSO zone	2 vars
BBK	BBK Federal Civil Protection	CONTINUOUS	Kreis	1 var
KBA	Kraftfahrt-Bundesamt (KBA)	QUARTERLY	Kreis	1 var
STDET	StartupDetector Germany	QUARTERLY	Kreis	1 var
BNA	Bundesnetzagentur (BNetzA)	ANNUAL	Kreis	8 vars
DESTA	Destatis (Federal Statistics)	ANNUAL	Kreis (AGS)	8 vars
UBA	Umweltbundesamt (UBA)	ANNUAL	Kreis / station	4 vars
JRC	JRC DSO Observatory	ANNUAL	DSO-level	2 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km + station	3 vars
EURO	Eurostat Energy Statistics	ANNUAL	NUTS2	3 vars
DESI	DESI Digital Economy Index	ANNUAL	EU Regional	2 vars
BMWK	BMWK Energy Policy Data	ANNUAL	Bundesland	2 vars
BMUV	BMUV Environmental Data	ANNUAL	Bundesland	1 var
BMF	BMF Fiscal Statistics	ANNUAL	Kreis	1 var
BFS	BfS Radiation Protection	ANNUAL	Bundesland	1 var
SOEP	SOEP Panel (DIW Berlin)	ANNUAL	Regional	2 vars
BDEW	BDEW Energy Association	ANNUAL	Bundesland	2 vars
INKAR	INKAR Regional Indicators	ANNUAL	Kreis	2 vars
BNETZA-M	BNetzA Monitoring Report	ANNUAL	DSO-level	2 vars
BBSR	BBSR Spatial Planning	ANNUAL	Kreis	1 var
IWH	IWH Halle Economic Research	ANNUAL	Bundesland	1 var
LAEA	Länderarbeitsgemeinschaft Energie	ANNUAL	Bundesland	1 var
BGR	BGR Geological Survey (WMS)	STATIC	Kreis	3 vars
D10	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars
DIM	Dimovski et al. (2025)	STATIC	Municipal	3 vars
IEEE-1	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars

GFZ	GFZ German Research Centre	STATIC	~5 km	2 vars
GKGZ	Gemeindekennziffern / AGS Registry	STATIC	Kreis (5-digit)	1 var
ISO-9223	ISO 9223 Corrosion	DERIVED	Kreis	1 var
SMARD	SMARD Energy Market Data	HOURLY	Bidding zone	3 vars
ENTSE	ENTSO-E Transparency	HOURLY	Bidding zone	2 vars
DWD	Deutscher Wetterdienst (DWD)	DAILY	~1 km	3 vars
D-OM	Open-Meteo / ERA5	HOURLY	~1 km	1 var

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Weekly (2 sources) — OSM diffs + weather reanalysis

OSM Overpass · Open-Meteo/ERA5

Monthly (3 sources) — Grid load + procurement data

MaStR Bulk · SMARD · TSOs

Quarterly (2 sources) — DER registry + business data + macro indicators

KBA · StartupDetector

Annual (20 sources) — Core grid quality + demographics + environment

BNetzA · Destatis · SOEP · BDEW · BMWK · BMUV · BMF · UBA · EEA · Eurostat · DESI · INKAR · BNetzA Monitoring · BBSR · IWH · LAEA · BfS · DWD · JRC · BBK

Static (8 sources) — Standards + scenarios + seismic hazard

Dimovski · IEEE/IEC/CIGRÉ · Copernicus CMIP6 · GFZ · BGR · ISO 9223 · GKGZ/AGS · ENTSO-E

C.2 Fleet Score Snapshot

FLEET SIZE
13,251
substations scored

MEDIAN R
0.507
P5=0.345 · P95=0.629

HIGH + CRITICAL
7040
53.1% of fleet

HIGH CONFIDENCE
100%
13251 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 0 (0.0%) · Medium 6211 (46.9%) · High 7040 (53.1%) · Critical 0 (0.0%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **13,251 substation scores remain unchanged** this edition. The fleet median R of 0.507 (mean 0.499) reflects a distribution skewed toward the lower bands, with 0.0% in Low and 46.9% in Medium. The first material score changes are expected when BNetzA publishes the annual SAIDI/SAIFI statistics (affecting C1–C4 and R6a) and when BNetzA updates the Marktstammdatenregister (MaStR) DER registry (affecting T1). High-frequency sources (OSM, Open-Meteo) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

17 changes tracked across P0, P1, and P2 releases

F1	NEW	New T component — Energy Transition Exposure (T1)	§2, §4
F2	NEW	New R6a — Restoration Speed Modifier (CAIDI-based)	§5
F3	ENHANCED	R3 enhanced — Energy Poverty Vulnerability (V_socio)	§5
F4	ENHANCED	R4 enhanced — Graph-theoretic betweenness + bridge detection	§5
F5	ENHANCED	R2 enhanced — Climate Trajectory (CMIP6 SSP2-4.5)	§5
F6	ENHANCED	R7 Digital Readiness — Kreis-level DESI model	§5
L1	NEW	New I5, I7–I9 metrics — thermal, load, corrosion, hydrogeo	§2, §8
L2	ENHANCED	E2 Innovation Enrichment — HRST + startup density	§6
L3	ENHANCED	V_socio Fiscal Enrichment — BMF + INKAR + energy price	§5
L4	ENHANCED	R3 Demographic Shift Amplifier — Destatis net migration	§5
L5	DATA	95/95 variables operational (100%). 35 data sources total.	§8, §12
G1	DATA	d02 MaStR upgraded to bulk SQLite download — Kreis-level DER registry	§12
G2	DATA	d08 BGR upgraded to WMS live API — Kreis-level geological data	§12
G3	DATA	d13 OSM upgraded to Overpass API — 9,051 real substations	§12
G4	NEW	R6b Network Topology modifier — centrality + ring analysis from OSM graph	§5
G5	NEW	Network & Topology layer (H): 7 variables — Netzentwicklungsplan, ring analysis	§12
G6	NEW	Output Scores layer (I): 7 variables — risk_score, ETTC, stationary probs	§12

SECTION D — MONTHLY DEEP-DIVE

The Schleswig-Holstein Wind Corridor: Where the Energy Transition Is Outrunning the Grid

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Schleswig-Holstein Wind Corridor · **002** Bayern DER saturation · **003** NRW Industrial Belt climate vulnerability · **004** Sachsen economic transformation · **005** Brandenburg transmission bottleneck · **006** Niedersachsen offshore wind integration · **007** Baden-Württemberg automotive electrification · **008** Hessen financial centre grid stress · **009** Mecklenburg-Vorpommern rural grid · **010** Thüringen green hydrogen corridor · **011** Hamburg/Bremen port electrification · **012** Year in review.

D.1 Why Schleswig-Holstein, Why Now

SH SUBSTATIONS

466

42 HV · 424 MV

HIGH BAND

406

87.1% of SH

CORRIDOR MEDIAN R

0.530

↑ above fleet median

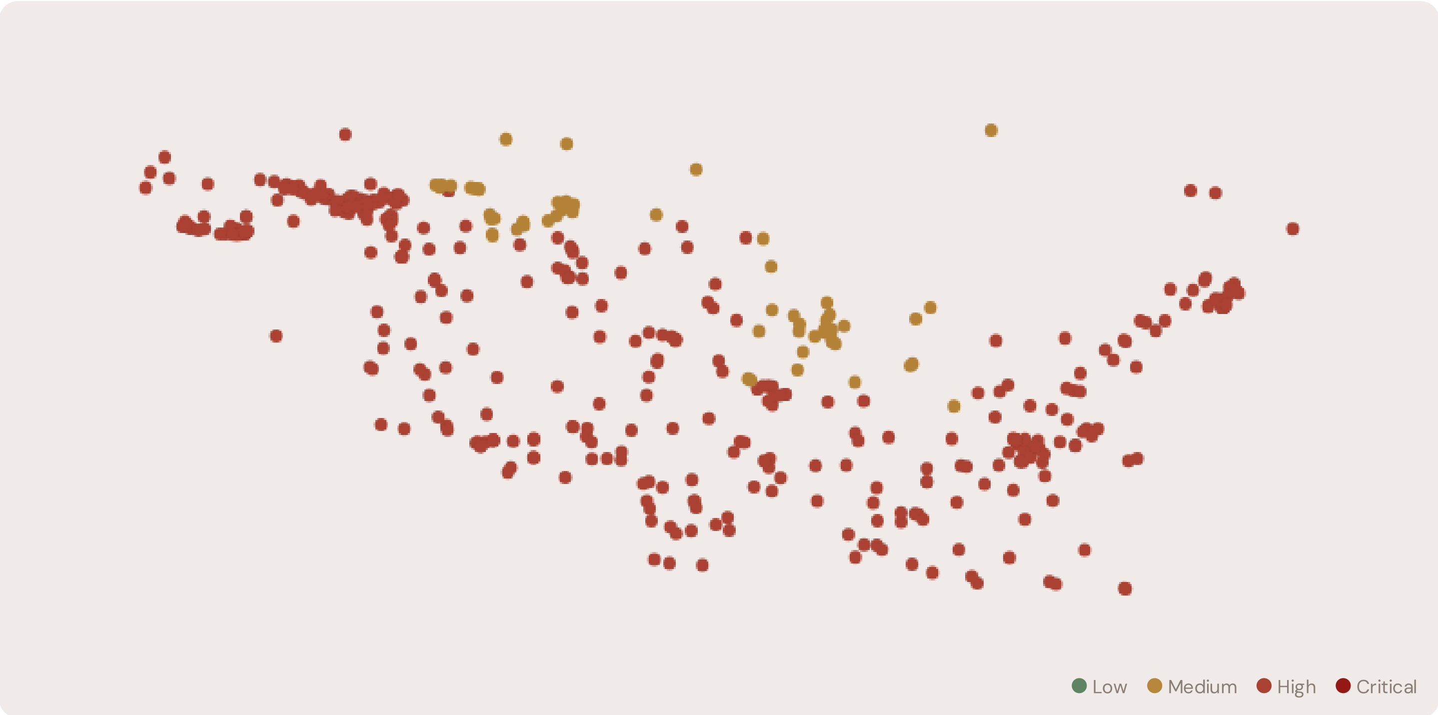
WORST KREIS

Dithmarschen

24 / 24 High/Critical

Schleswig-Holstein (SH) is Germany's premier wind-energy Bundesland, with over 4,000 MW of installed capacity. However, this renewable abundance creates grid stress: high S (saturation) and T (transition) components indicate congestion where wind generation outpaces grid absorption and curtailment services. The corridor median R of 0.530 (above fleet) reflects this transition tension. 406 substations in SH are in the High or Critical risk bands, concentrated in the northern coastal Kreise where offshore wind interconnects feed into aging onshore grids.

D.2 Corridor Map and Scoring



SUBSTATION	KREIS	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Heide	Dithmarschen	0.594	High	0.552	0.251	0.502	0.224	0.383	0.534	
Umspannwerk Marne-West	Dithmarschen	0.594	High	0.552	0.251	0.502	0.224	0.383	0.534	
Substation 165625451	Dithmarschen	0.594	High	0.552	0.251	0.502	0.224	0.383	0.534	
Umspannwerk Strübbel	Dithmarschen	0.594	High	0.552	0.251	0.502	0.224	0.383	0.534	

SUBSTATION	KREIS	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Tönning	Dithmarschen	0.593	High	0.552	0.251	0.502	0.224	0.383	0.534	
Umspannwerk Quickborn Dithmarschen	Dithmarschen	0.593	High	0.552	0.251	0.502	0.224	0.383	0.534	
Friedrichstadt	Dithmarschen	0.592	High	0.552	0.251	0.502	0.224	0.383	0.534	
Dieksanderkoog	Dithmarschen	0.592	High	0.552	0.251	0.502	0.224	0.383	0.534	
Meldorf	Dithmarschen	0.591	High	0.552	0.251	0.502	0.224	0.383	0.534	
Ostermoor/Ost	Dithmarschen	0.591	High	0.552	0.251	0.502	0.224	0.383	0.534	
Umspannwerk Süderdonn	Dithmarschen	0.590	High	0.552	0.251	0.502	0.224	0.383	0.534	
Umspannwerk Brunsbüttel (TenneT)	Dithmarschen	0.590	High	0.552	0.251	0.502	0.224	0.383	0.534	
Ostermoor-West	Dithmarschen	0.590	High	0.552	0.251	0.502	0.224	0.383	0.534	
UW Nortorf	Rendsburg-Eckernförde	0.590	High	0.552	0.173	0.503	0.224	0.359	0.503	
Umspannwerk Heide-West	Dithmarschen	0.590	High	0.552	0.251	0.502	0.224	0.383	0.534	
Linden	Dithmarschen	0.589	High	0.552	0.251	0.502	0.224	0.383	0.534	
Umspannwerk Katenstedt	Rendsburg-Eckernförde	0.589	High	0.552	0.173	0.503	0.224	0.359	0.503	
Pollhorn	Rendsburg-Eckernförde	0.589	High	0.552	0.173	0.503	0.224	0.359	0.503	
Umspannwerk Kernkraftwerk Brunsbüttel	Dithmarschen	0.588	High	0.552	0.251	0.502	0.224	0.383	0.534	
Wöhrden	Dithmarschen	0.588	High	0.552	0.251	0.502	0.224	0.383	0.534	

D.3 Component Analysis

COMPONENT PROFILE

Continuity 0.552 / 0.30 (184%)

Voltage 0.164 / 0.10 (164%)

MODIFIER IMPACT

R3 Consequence 1.081 ↑

R4 Topology 1.146 ↑

Infrastructure	0.496 / 0.25 (198%)
Economic	0.216 / 0.10 (216%)
Saturation	0.347 / 0.20 (173%)
Transition	0.499 / 0.05 (999%)

R6a Restoration	1.011 ↑
R6b Network	1.000 →
R7 Digital	0.997 →

SH's component profile differs sharply from the fleet average. Continuity (C) is elevated at 0.552 (184% of ceiling), reflecting SAIDI pressures from wind curtailment and frequency instability. Saturation (S) at 0.347 and Transition (T) at 0.499 are among Germany's highest, driven by the mismatch between renewable injection and demand-following flexibility. Modifiers show R7 (Digital Readiness) averaging 0.997 — above the German average, reflecting SH's investment in SCADA and grid management. However, R6b (Network Topology) at 1.000 indicates network stress concentration in key transmission nodes.

D.4 Kreis Breakdown

Dithmarschen	24 / 24 (100%)
Rendsburg-Eckernförde	20 / 20 (100%)
Steinburg	25 / 25 (100%)
Herzogtum Lauenburg	16 / 16 (100%)
Segeberg	29 / 29 (100%)
Stormarn	31 / 31 (100%)
Nordfriesland	150 / 150 (100%)
Pinneberg	21 / 21 (100%)
Schleswig-Flensburg	18 / 18 (100%)

Ostholstein 72 / 72 (100%)

Plön 0 / 6 (0%)

Flensburg 0 / 29 (0%)

Kiel 0 / 25 (0%)

Within Schleswig-Holstein, grid risk concentrates in Dithmarschen (24 High/Critical substations). This coastal Kreis hosts the main offshore wind export corridors and faces the steepest ramp rates from wind variability. The top 5 Kreise account for 114 of 406 High/Critical substations.

D.5 Implications

The Schleswig-Holstein Wind Corridor reveals a fundamental challenge for Germany's Energiewende: renewable generation capacity has grown faster than grid infrastructure to absorb it. Grid operators face curtailment decisions that constrain returns on wind assets — a situation the T component captures explicitly. Battery energy storage (BESS) at these high-risk, high-saturation corridors has asymmetric value: it avoids curtailment losses while providing local voltage support and inertia. SSI valuations of BESS deployment at SH substations will quantify this optionality, separating locations where storage unlocks value from those where grid reinforcement is the better path.

SECTION E

European Context Card

Why Europe matters: Germany does not operate in isolation. SAIDI (System Average Interruption Duration Index) is the standard reliability metric across the EU, enabling direct comparison. Germany's 11.7 minutes ranks best in Europe — yet the SSI reveals hidden vulnerability concentrated in specific regions where transition stress outpaces legacy infrastructure. Europe's ACER resilience framework depends on this asset-level view.

E.1 SAIDI Comparison: European Benchmark

GERMANY'S POSITION

11.7 min

Lowest SAIDI in EU — but SSI shows concentrated risk pockets

EU PEER CONTEXT

Germany	11.7
Denmark	24.3
Netherlands	28.1
France	61.5

Italy	36.8

This month's key insight: The Schleswig-Holstein corridor deep-dive shows **466 substations** (of 466) with S-component above 70% of its weight ceiling — indicating saturation-of-supply performance consistent with Germany's wind transition tension. Schleswig-Holstein's average S component is 0.347 compared to the German fleet average of 0.341 — a ratio of 1.0× — making it the grid's saturation hotspot. The Continuity component tells the inverse story: despite being Germany's renewable leader, SH's SAIDI-proxy reflects not equipment failure but renewable variability-driven instability. This distinction — where transition stress masquerades as traditional reliability risk — is precisely what the SSI's modular component design captures.

Regional Risk Ranking — All 16 German Bundesländer

Where does Schleswig-Holstein sit within the national landscape? Bundesländer sorted by mean R_median, with band distribution and fleet comparison.

Sorted by mean R_median (highest risk first)

1	Mecklenburg-Vorpommern	0.652	201 HC
2	Thüringen	0.633	506 HC
3	Sachsen-Anhalt	0.626	335 HC
4	Sachsen	0.615	545 HC
5	Brandenburg	0.601	671 HC
6	Bayern	0.557	3481 HC
7	Schleswig-Holstein	0.526	406 HC
8	Rheinland-Pfalz	0.506	574 HC
9	Niedersachsen	0.484	321 HC
10	Berlin	0.435	0 HC
11	Baden-Württemberg	0.409	0 HC
12	Nordrhein-Westfalen	0.370	0 HC
13	Saarland	0.355	0 HC
14	Hamburg	0.349	0 HC
15	Hessen	0.349	0 HC
16	Bremen	0.333	0 HC

Bundesland-level risk ranking reveals sharp disparities across Germany's 16 regions. The highest-risk Bundesländer (Mecklenburg-Vorpommern (0.652), Thüringen (0.633), Sachsen-Anhalt (0.626)) combine industrial infrastructure, aging transmission systems, and complex interconnection with high-capacity renewable injection sites. Schleswig-Holstein ranks 7th nationally by mean R_median (0.526), with 406 of its 466 substations in the High/Critical bands. However, this ranking masks the underlying cause: SH's risk is driven by transition stress (S + T components), not by traditional infrastructure age (I component). Eastern Bundesländer (Sachsen, Thüringen, Brandenburg) show elevated risk driven by both aging infrastructure and economic transition factors. The SSI's component decomposition allows policymakers to distinguish grid reinforcement needs from transition management — a critical input to Energiewende planning.

SSI Enhanced Neural Network Architecture

The ENN framework: The SSI's risk scores flow from a 5-layer neural network architecture: (1) Data — aggregation and normalisation of 95 variables from 35 public sources into component sub-metrics; (2) Valuation — conversion of grid metrics into NPV using real-options and Monte Carlo; (3) Neural Net — gradient-boosted tree ensemble + DNN quantile regression to predict valuations and uncertainty; (4) Coverage — country-by-country calibration and registry expansion; (5) Platform — product delivery (Dashboard, API, Value-Add Cards, Advisory). This section explores Layer 3, the predictive engine, and its grounding in German grid data.

Layer 1 — Data	Layer 2 — Valuation	Layer 3 — Neural Net	Layer 4 — Coverage	Layer 5 — Platform
SSI v4.0 component taxonomy, normalisation, MC engine	DNO + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	Country prioritisation, DCI, registry, calibration	Products, API, revenue model, unit economics

LAYER 2 §0.4.1 · edition · May 2026

Valuation Stack — DNO + Community Value

BESS deployment creates value on two dimensions

A battery energy storage system (BESS) deployed at a substation generates returns through two distinct channels: (1) DNO value — deferral of grid reinforcement, reduced curtailment, arbitrage from price spreads, frequency response services, and avoided transformer replacement; (2) Community value — reduced outage duration, avoided economic loss (VoLL), and emissions reduction. The SSI models both using Monte Carlo: DNO value is stochastic because market prices and utilization rates fluctuate; community value is stochastic because outage probability and duration are uncertain (reflected in CI_width). The combined NPV distribution reveals whether BESS is investment-grade. German DNOs benefit from regulated asset base recovery mechanisms, making BESS NPV less volatile than in merchant markets. However, community value — avoided industrial disruption in manufacturing-dense Bundesländer, avoided social cost of residential outages in high-poverty regions — creates the case for public subsidy.

KEY CONCEPTS

- → 2-stack valuation: DNO (grid deferral, arbitrage, services) + Community (VoLL avoidance, emissions)
- → Monte Carlo propagates DNO price + utilization uncertainty
- → Community value quantifies social cost of outage duration
- → German regulated framework: NPV more stable than merchant markets

LIVE SSI DATA CONNECTION

Across the 13,251-substation German fleet, the average consequence multiplier (R3) is 1.124, indicating significant socio-economic exposure. 7040 substations (53.1%) are classified High or Critical — these are the highest-priority candidates for BESS deployment where DNO and community value stacks are largest.

Coming next month: LAYER 5 Platform — Product Delivery Pipeline — *From raw analytical outputs to four commercial product formats*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 003

MV Rural Grid

Deep-dive into Mecklenburg-Vorpommern's grid risk profile — substation map, component analysis, Kreis breakdown. European Context Card: Transition stress in low-density network.

NEXT DATA REFRESH

Q3 2026 — 2 Quarterly Sources

KBA, StartupDetector are due for refresh in Q3. Impact: DER registry updates (T1), business fabric indicators (E2), macro-economic data. Key upcoming: BNetzA SAIDI/SAIFI refresh and MaStR updates.

FLEET SPOTLIGHT

3141 Substations: Low GDP + High Risk

3141 substations sit in below-median GDP regions while scoring High/Critical for grid risk. These are economically vulnerable areas with the least fiscal capacity for grid investment. Federal Energiewende support needs to prioritize these locations.

Edition 003 will be published on the second Thursday of June 2026. Deep-dive region: **Mecklenburg-Vorpommern**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a German utility, Stadtwerk, Bundesnetzagentur, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 002

Published 14 May 2026 · SSI Index v4.0.2 · 13,251 substations · 61,625 grid lines · 6 components · 20 metrics · 5 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (3.76B operations) · CMIP6 SSP2-4.5